Computational and Theoretical Properties of Neural Coordinate Transformations

Coordinate transformations underlie the diversity of spatial coding schemes observed throughout the brain. For example, recent studies have shown retrosplenial cortex (RSP) encodes egocentric variables (measuring features in the world with respect to the animal's state) alongside allocentric ones (measuring the animal's state with respect to features in the world)[1]. Previous studies have examined how neural circuitry may generate these codes[2] and transition between them[3], however a normative, mechanistic understanding of the structure and dynamics of circuits involved in coordinate transformation is lacking. Presently, we investigate a Recurrent Neural Network (RNN) circuit model to address these gaps.

We trained an RNN to perform a coordinate transformation relevant to mouse's scape behavior: given the two allocentric variables of head direction and shelter position, the network computes the angle between them, the egocentric 'head-shelter angle'. It then maintains the egocentric coordinate from a noisy angular velocity signal. By examining the functional cell types which emerge in the trained network, we recover behaviour consistent both with retrosdplenial cortex data [1] and with continuous attractor network models maintaining a variable by integration of a noisy signal[4], suggesting the network uses well-characterised dynamics to operate within the transformed space. To understand the initial computation of this transformation, we examine the network dynamics driving the transition from one coordinate representation to another. We characterise these representations using their intrinsic geometry within the state of the trained RNN[5]. Overall, this simplified example of coordinate transformation illustrates a mechanistic picture of neural coordinate transformations as dynamical transitions between continuous attractors, and provides normative support for this interpretation by the learned behaviour of task-optimised RNNs.

Additional details: We discuss coordinate transformation in the terms of Campagner et al.[1]. Given two allocentric signals, head direction (θ_1 below) and shelter position (θ_2), measured from an external landmark (e.g., 'due east' or the positive x-axis), the network must produce an egocentric signal, the head-shelter angle $\Delta\theta = \theta_2 - \theta_1$. We only consider rotation of the mouse about a point; that is, the transformation can be described entirely in terms of angles. Abstractly, one can think of the transformation as requiring taking two coordinates in one coordinate system, and transforming one coordinate into a new coordinate system whose zero is given by the other input; crucially, the interaction between coordinates is non-linear.

Model: We employ an RNN model adapted from[2]. The network state $\mathbf{x} \in \mathbb{R}^N$ has dynamics given by $\tau\dot{\mathbf{x}} = -\mathbf{x}(t) + P\phi(\mathbf{x}(t)) + B\mathbf{u}(t) + \mathbf{b} + \eta(t)$. In the present task, $\mathbf{u}(t)$ is structured to represent the desired coordinate-transformation computation. For each trial, we sample θ_1 and θ_2 uniformly in $[-\pi, \pi)$, corresponding to the head-direction and shelter-position at the beginning of an experimental trial. We then simulate angular velocity of the head, $\omega(t)$ by random walk. Concretely, for $t < \tau$, four inputs are given: $\sin\theta_1$, $\cos\theta_1$, $\sin\theta_2$, $\cos\theta_2$; this is the period where the network is presumed to initialise its representation and compute the coordinate transformation. For $t \geq \tau$, only $\omega(t)$ is given, which is the period where continuous-attractor characteristics are observed. We train the RNN to perform the coordinate transformation by a linear readout of firing rates: $\mathbf{z}(t) = W\phi(\mathbf{x}(t)) + \mathbf{c}$. We prescribe a target output signal $\mathbf{y}(t)$ as $\langle \sin\Delta\theta(t), \cos\Delta\theta(t)\rangle$, where $\Delta\theta(t) = (\theta_2 - \theta_1) + \int_0^t \omega(t')dt'$. For all times after the allocentric inputs are removed, i.e. $t \geq \tau$, we define a loss function as $\mathscr{L} = \langle ||\mathbf{y}(t) - \mathbf{z}(t)||^2\rangle + \lambda_1\langle\phi(\mathbf{x}(t))\rangle + \lambda_2\langle||P||^2\rangle$, and train by gradient descent on the loss function with respect to P, B, W, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{x}(0)$.

Single unit analysis: We calculate the average firing rate of all units concurrent with each value of θ_1, θ_2, $\Delta\theta(t)$, and $\omega(t)$. The tuning curves for $\Delta\theta(t)$ were well approximated by cosine curves, and for $\omega(t)$ by a line; we fit these curves for each neuron in the network and used the amplitude of the cosine curve, and slope of the line, as measures of strength of tuning for $\Delta\theta$ and ω, respectively. We calculated the average synapse from $\Delta\theta$-tuned neurons by averaging synapses in the network, weighted by the strength of tuning of the pre-synaptic neuron to $\Delta\theta$, and equivalently for ω.

PCA analysis: For analysis of the transformation computation, the angular velocity-integration aspect of the task was removed at test-time by setting $\omega(t) = 0$. We performed a PCA decomposition on the network state across test trials for each timestep. We visualise the cumulative sum of leading eigenvalues to reveal a collapse from a four-dimensional subspace of activity onto a two-dimensional subspace of activity at allocentric input offset.

Metric analysis: To quantify the geometry of network state in the input- and output-determined subspaces identified by PCA analysis, we calculate the metric of the network state. The network state is considered to be parametrised by the allocentric inputs seen at the beginning of a trial: $\mathbf{x}(\theta_1, \theta_2, t) = \mathbf{x}(0) + \int_0^t \dot{\mathbf{x}}(\theta_1, \theta_2, t')dt'$. The derivative of this state with respect to the task variables gives a measure of the direction in state space that a small change in the inputs would produce after time t; the dot-product between these derivatives measures the magnitude of this change, or the direction between changes in different task variables. We estimate these derivatives by finite differences (evaluated across the range of θ_1 and θ_2, and stack their dot-products into one matrix. We observed that the metric on the network at early timesteps resembles the metric on the inputs passed through an untrained network–that is, the geometry is initially dominated by the inputs, rather than the structure of the network. Similarly, at later timesteps, the geometry resembles that of a flat ring in two linear dimensions, where position on the ring is parametrised by $\Delta\theta$–that is, the geometry becomes dominated by that of the trained outputs.

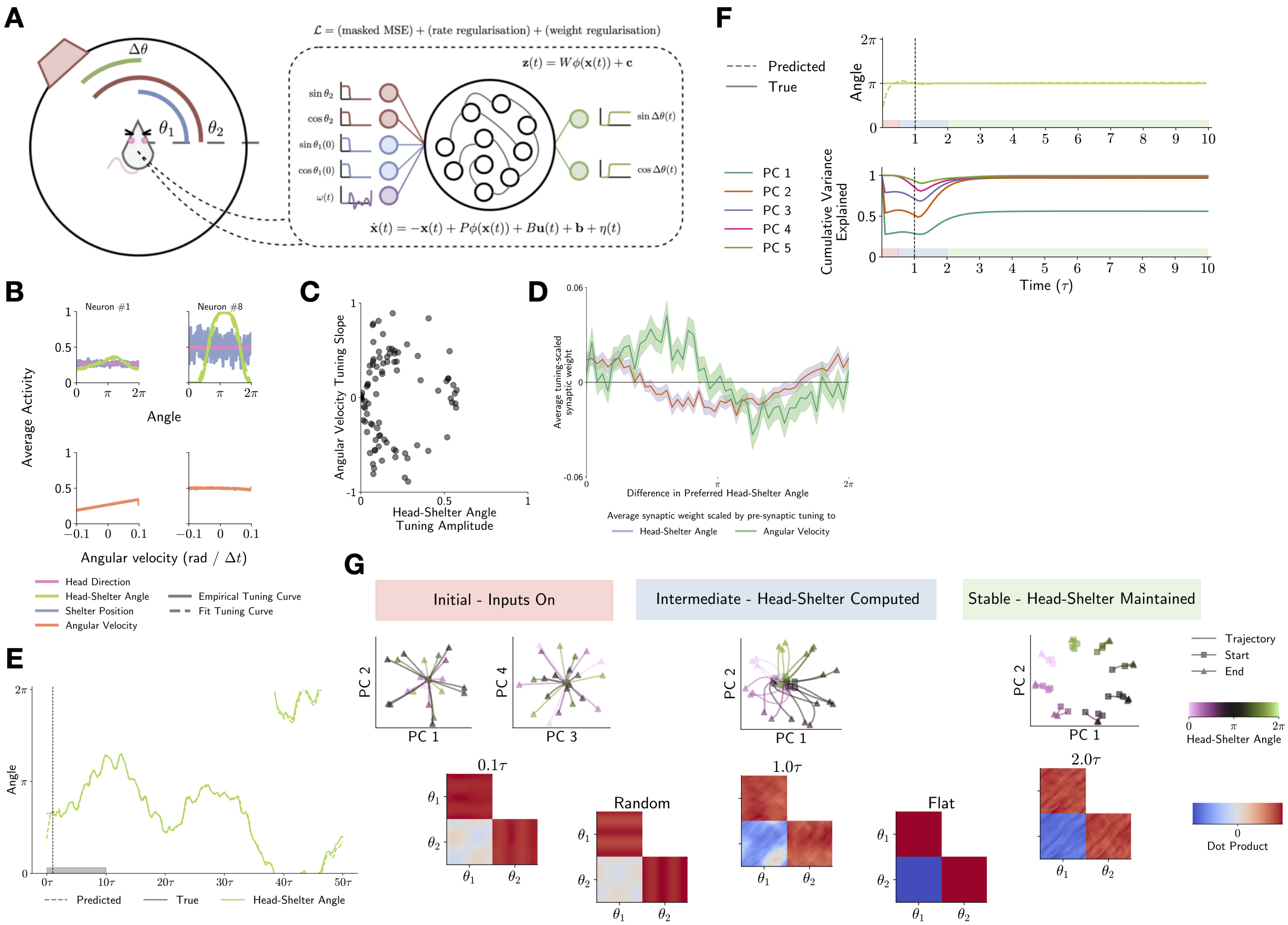

Figure 1 **A)** Schematic of RNN model of coordinate transformation between allocentric measures θ_1 and θ_2, and the egocentric $\Delta\theta$. **B)** Example of single-neuron tuning to task variables. Tuning curves for head-shelter angle ($\Delta\theta$) and angular velocity (ω) are well approximated by cosine and linear functions, respectively. **C)** Tuning to head-shelter angle and angular velocity distribute across the population, coarsely stratifying 'compass' and 'shifter' units. **D)** Average synaptic weight between neurons scaled by pre-synaptic tuning reveals classical continuous actractor properties, where the head-shelter angle is maintained by symmetric or 'Mexican hat' connectivity, and adjusted in response to angular velocity by asymmetric connectivity. **E)** Network output in one example training trial, showing the network learns the transformation well. Dashed line denotes offset of allocentric inputs, and onset of angular velocity input. **F)** *(top)* A testing-time task is introduced without angular velocity to isolate coordinate transformation computation. *(bottom)* PCA on the network state in testing task reveals collapse from a 4-dimensional (input dominated) to 2-dimensional (output-dominated) subspace of activity at offset of allocentric inputs. Note that by the time activity registers as 2-dimensional, the network has already found the head-shelter angle; this collapse is where the transformation occurs. **G)** Testing-trial in F is subdivided into initial, intermediate, and stable periods. *(top)* Visualising network trajectories in PC space reveals collapse from structure where head-shelter angle is undifferentiated (initial period) to a ring where head-shelter angle is well separated (stable); dynamics of the intermediate period correspond to dynamics of the transformation between these two representations. *(bottom)* Metric on network representations in each state reveals initially (initial period, 0.1τ), network geometry resembles that of an untrained network (Random)–that is, network structure is dominated by inputs. Later (stable period, 2.0τ) geometry resembles a flat ring encoding $\Delta\theta$ (Flat). Intermediate period geometry shows interpolation between representations.

References [1] Campagner et. al,2023 [2] Cueva & Wei, 2018 [3] Byrne & Becker, 2008 [4] Khona & Fiete, 2022 [5] Hausser & Ray, 2017

1. Dario Campagner, Ruben Vale, Yu Lin Tan, Panagiota Iordanidou, Oriol Pavón Arocas, Federico Claudi, A. Vanessa Stempel, Sepiedeh Keshavarzi, Rasmus S. Petersen, Troy W. Margrie, and Tiago Branco. A cortico-collicular circuit for orienting to shelter during escape. Nature, 613(7942):111–119, January 2023.
2. Christopher J. Cueva and Xue-Xin Wei. Emergence of grid-like representations by training recurrent neural networks to perform spatial localization, March 2018. arXiv:1803.07770 [q-bio].
3. Patrick Byrne and Suzanna Becker. A principle for learning egocentric-allocentric transformation. Neural Computation, 20:709–737, 03 2008.
4. Mikail Khona and Ila R. Fiete. Attractor and integrator networks in the brain. Nature Reviews Neuroscience, 23:744–766, 12 2022.
5. Michael Hauser and Asok Ray. Principles of riemannian geometry in neural networks, 2017.